

[7] GoblinX mini

Copy *vmlinux* and *initrd.gz* of GoblinX from */mnt/tmp7/boot* into *mbcd/boot/goblinx*.

```
$ mkdir mbcd/boot/goblinx
$ cp /mnt/tmp7/boot/vmlinux \
/mnt/tmp7/boot/initrd.gz mbcd/boot/goblinx
```

Then copy the */mnt/tmp7/goblinx* directory into *mbcd/*.

```
$ cp -r /mnt/tmp7/goblinx mbcd/
```

This ends the copying process of our seven operating systems.

Phase III: Creating the boot directory

Now we will install the boot files that are required to make the ISO file or the USB bootable. We can use either the GRUB or SYSLINUX bootloader and we will cover the installation of both in this article. In Phase V we will either use GRUB or SYSLINUX to make the CD ISO image bootable (though both the bootloader files can reside in the image without conflicts).

[1] Installing GRUB

Make a directory named *grub* under *mbcd/boot/*.

```
$ mkdir mbcd/boot/grub
```

To boot from the CD, Grub needs a special image for loading Stage 2. You need the *stage2_eltorito* file. If you have got Grub installed in your system, then execute any one of the following commands and note the path:

```
$ locate stage2_eltorito
```

...OR

```
$ rpm grub rpm -q grub | grep stage2_eltorito
```

Else, download the GRUB source, compile it and get the *eltorito* image.

I am assuming you have the *stage2_eltorito* under the */eltorito_dir* which represents the absolute path of the location of the *stage2_eltorito* file in your system. Copy *stage2_eltorito* into *mbcd/boot/grub/*.

```
$ cp /eltorito_dir/stage2_eltorito mbcd/boot/grub/
```



Note: The Grub files that have to be copied in order to boot from the USB drive are described in Phase VI, Section [2].

[2] Installing SYSLINUX

Two files are required to install the SYSLINUX: *isolinux.bin* is needed to make the image bootable, and *vesamenu.c32*

to enable the graphical menu system.

If you had SYSLINUX installed previously, then run one of the following commands and note down the paths:

```
$ locate vesamenu.c32 isolinux.bin
```

...OR

```
$ rpm -q syslinux | grep vesamenu.c32 && rpm -q syslinux | grep isolinux.bin
```

If you do not have SYSLINUX, install it using your package manager. Or download the source from the SYSLINUX website, extract and run *make*, but don't install it (if you want to avoid installation—that is, only execute */configure && make*). Get the required files from the source directory.

Make a directory called *mbcd/boot/isolinux*, and copy *isolinux.bin* and *vesamenu.c32* to *mbcd/boot/isolinux*.

```
$ mkdir mbcd/boot/isolinux
```

```
$ cp /path_to/vesamenu.c32 \
/path_to/isolinux.bin mbcd/boot/isolinux
```



Shortcut: To avoid downloading and compiling SYSLINUX or Grub, just copy *isolinux.bin* and *vesamenu.c32* for SYSLINUX and *stage2_eltorito* file from any distros */boot* directory, and continue to the next step.

Phase IV: Making the configuration file

The previous phase sets the bootloader files; now we need to instruct the bootloader which OS to boot. We also need a menu from where the user can select an OS to boot and we need a configuration file. The Syntax differs in Grub and SYSLINUX. To know more about config file syntax, use the man pages.

Some distros need different boot parameters to be passed for USB and CD versions. We will present the configuration files for the live CD version in Section 1 (GRUB) and 2 (ISOLINUX) below, after which we will describe the modifications needed for the live USB version in Section 3.

[1] The Grub configuration file

The Grub configuration file will have three lines for each OS entry. The first line, "label text", displays the entry that we see on a Grub menu. In the second line we specify the location of the OS kernel to be loaded, and pass the needed boot parameters. In the third line, we point the initial ramdisk through the *initrd* command associated with the kernel.

You can add more than one entry for each distro, each using an OS-specific boot parameter. You can also write the parameters in a text file and display it when some key is pressed or use other ways to display text parameters. Check